DITSU KUNDU

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PROFESSIONAL SUMMARY

Final-year Electronics & Communication Engineering undergraduate with strong grounding in data structures, algorithms, and object-oriented design, applied through full-stack and AI-integrated systems projects. Hackathon winner, MLH mentor/organizer and ServiceNow Certified System Administrator.

INTERNSHIP EXPERIENCE

Generative AI Intern | IEM UEM Generative AI COE, Kolkata, India

September 2025 - June 2026

- Built an AI-powered SQL Query Generator in an agile team, converting natural language to SQL via a Qwen2.5 + Gemini pipeline with feedback-driven regeneration

VLSI Trainee | Jadavpur University, Kolkata

May 2025 - July 2025

- Simulated 15+ digital VLSI circuits in Xilinx Vivado and verified results using test benches
- Programmed an Artix-7 FPGA using Verilog to validate designs in real time

EDUCATION

B. Tech, Electronics & Communication Engineering, Institute of Engineering & Management (IEM), Salt Lake / University of Engineering & Management (UEM), Kolkata *July 2023 - Expected May 2027*

- **GPA: 8.62/10**
- **Relevant Coursework:** Data Structures & Algorithms | Operating Systems | DBMS | Object-Oriented Programming | Cloud Computing | Digital Electronics

Class XII (ISC, PCM + Computer Science) - 86.33%, St. Augustine's Day School *May 2021 - May 2023*
Class X (ICSE) - 86.9%, St. Claret School *May 2021*

PROJECTS

Queryra – AI Powered Data Querying Platform

- Created a full-stack application and a standalone command-line interface which transforms natural-language questions into validated, read-only SQL queries by using Gemini and a schema-aware generation pipeline.
- Integrated CSV schema discovery, SQL safety validation, and local DuckDB execution to provide query results, enabling command-line dataset selection and support for user-provided Gemini API keys.

"Watt-Watch Phantom Cut" Intelligent Energy Auditor (Hackathon Winner, 50+ teams)

- Engineered a real-time electrical device monitoring system using YOLOv8, detecting 10+ devices and triggering automated power-wastage alerts via ESP32
- **GitHub**: github.com/LeonDae/WattWatch_PhantomCut

TermShield – AI Contract Risk Analyzer for Freelancers

- Created a full-stack application and standalone command-line interface that transforms natural-language questions into validated, read-only SQL queries using Gemini and a schema-aware generation pipeline.
- Implemented RAG with embeddings and vector search to ground analysis in reference contract templates, reducing reliance on the model alone

SKILLS

Languages: Java (OOP) | Python | SQL
Core CS: Data Structures & Algorithms (80+ problems - LeetCode) | OOP | Operating Systems | DBMS
Cloud: AWS (S3 - bucket versioning, static website hosting)
Frameworks & Tools: LangChain | FastAPI | Prompt Engineering | GitHub

ACHIEVEMENTS

- **Winner - Frost Hacks Season 2 (evaluated by TCS)**
- **Event Organizer - Diversion 2K26 (MLH Hackathon):** Led technical operations and engagement for 100+ teams, 300+ participants
- **Mentor - HackNex Season 2 (MLH Hackathon):** Guided 110+ teams, 400+ participants
- **Runner-up - ISMAW Sustainability Hackathon**

CERTIFICATIONS

- IBM - AI Agents using RAG and LangChain
- **ServiceNow - Certified System Administrator (CSA)**
- System Design Fundamentals by Pearson (In Progress)